

The empirical recurrence for OEIS A260283, recovered from its tail

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Abstract

OEIS A260283 records “Empirical recurrence of order 75” and keeps the recurrence itself in a linked file rather than in the entry, so the conjecture cannot be read off the entry at all. It can still be settled. The entry counts arrays under a condition local to a bounded window of lines, which makes the count a walk count on a finite digraph and forces a monic integer recurrence of order at most the number $S' = 231$ of states with distinct futures. Berlekamp–Massey on enough exact terms therefore returns the minimal such recurrence, and on the tail beginning at $n = 7$, so the recurrence is proved for every $n \geq 82$. Its last coefficient is $c_{75} = 1189085184 \neq 0$, so no further tail satisfies anything shorter; any order-75 recurrence the sequence satisfies from any point on must then have the same characteristic polynomial. The entry’s recurrence is the one displayed here, whatever its linked file contains, and it is proved.

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1 A conjecture that is not written down

OEIS A260283 is “Number of $(n+2) \times (7+2)$ 0..1 arrays with each 3×3 subblock having clockwise perimeter pattern 00000000 00000001 or 00001011.”. It has offset 1 and begins

1428, 6351, 42175, 192277, 984644, 5588466, 28517537, 145143631, ...

The entry states:

Empirical recurrence of order 75 (see link above)

As of the “Last modified” line on the live entry (Jun 02 2025 12:04:54, revision 6) this is still recorded as empirical, and it is still open. The coefficients are nowhere in the entry text: what is conjectured is only that a recurrence of that order exists.

2 The count is a walk count

The entry’s condition is local. It constrains a bounded window of consecutive lines of the array and nothing beyond it, so the admissible configurations of one window are the vertices of a finite digraph, an edge joins u to v when v may follow u , and an admissible array is exactly a walk. Writing M for the adjacency matrix and ι, τ for the indicator vectors of the admissible first and last windows,

$$a(n) = \iota^\top M^{j(n)} \tau$$

for a linear reindexing j fixed by the entry’s shape. States with identical futures contribute identically to that product and may be merged without changing any count; after merging $S' = 231$ states remain, and it is that bound every computation below uses.

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 231$.*

Proof. By the Cayley–Hamilton theorem $M^{S'}$ is an integer combination of $I, M, \dots, M^{S'-1}$, and multiplying that relation on the left by $\iota^\top M^j$ and on the right by τ turns it into a relation among $S' + 1$ consecutive values of a . $\square$

3 Why the question has to be asked of a tail

Running Berlekamp–Massey on the sequence from its first term returns an order LARGER than 75: the opening terms do not satisfy the recurrence, and a recurrence that holds only eventually always looks longer when the exceptional terms are included. That is not evidence against the entry. An OEIS empirical recurrence is asserted for n beyond some index, not from the first term, so the question has to be asked of the tails. It was asked of each tail in turn, and the tail beginning at $n = 7$ is the first whose minimal order is exactly the stated 75.

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S' is determined, together with its minimal recurrence, by any $2S'$ consecutive terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Applied to $2S' = 462$ exact terms from $n = 7$ on, it returns order 75 — the order the entry states — with integer coefficients:

$$a(n) = \sum_{i=1}^{75} c_i a(n-i),$$

c_1	1	c_{20}	−1470963946	c_{39}	16286429507240	c_{58}	3731349532928
c_2	0	c_{21}	8829771780	c_{40}	15454750692120	c_{59}	−69824109568
c_3	162	c_{22}	16853050011	c_{41}	6124862449392	c_{60}	−8962857841664
c_4	91	c_{23}	11430063453	c_{42}	−28291529358768	c_{61}	−1264871864320
c_5	111	c_{24}	−53854469492	c_{43}	−22297811627956	c_{62}	310205067264
c_6	−9421	c_{25}	−97185496910	c_{44}	−7580109233624	c_{63}	3215666055168
c_7	−13189	c_{26}	−65491498022	c_{45}	39883470579528	c_{64}	329000042496
c_8	−9330	c_{27}	255576985702	c_{46}	25577340001184	c_{65}	−217430310912
c_9	309822	c_{28}	427608625158	c_{47}	7503821715800	c_{66}	−908193251328
c_{10}	568064	c_{29}	278551790264	c_{48}	−45577586656416	c_{67}	−88321474560
c_{11}	394026	c_{30}	−968137390142	c_{49}	−22866277888400	c_{68}	65597276160
c_{12}	−6729890	c_{31}	−1462605482760	c_{50}	−5557266959616	c_{69}	188994355200
c_{13}	−13454328	c_{32}	−887512932782	c_{51}	42374153873632	c_{70}	25421414400
c_{14}	−9055668	c_{33}	2996597365196	c_{52}	15842522529984	c_{71}	−5491261440
c_{15}	101317060	c_{34}	3965255919174	c_{53}	2801925636544	c_{72}	−22281191424
c_{16}	205436500	c_{35}	2147897440236	c_{54}	−31982114764544	c_{73}	−2604662784
c_{17}	137714066	c_{36}	−7679645404148	c_{55}	−8613203048960	c_{74}	0
c_{18}	−1099546680	c_{37}	−8675121733764	c_{56}	−818260514432	c_{75}	1189085184
c_{19}	−2182542080	c_{38}	−4046257705976	c_{57}	19240794737664		

4 The recurrence, and the identification

Theorem 3. *The recurrence above holds for every $n \geq 82$, and it is the recurrence OEIS A260283 records.*

Proof. That it holds is Lemma 2 applied to the tail from $n = 7$, whose minimal recurrence it is.

For the identification, write $q_{\min}$ for the characteristic polynomial of that minimal recurrence, $x^{75} - c_1 x^{74} - \dots - c_{75}$. Its constant term is $-c_{75} = -1189085184$, which is not zero, so $q_{\min}(0) \neq 0$ and $q_{\min}$ has no root at the origin. A solution of a constant-coefficient recurrence

is a combination of terms $n^k \lambda^n$ with λ a root of its characteristic polynomial, and only the components with $\lambda = 0$ vanish after finitely many terms. Since $q_{\min}$ has none, no component of a dies out, and the minimal recurrence of every further tail is again $q_{\min}$ — the order cannot drop below 75 at any later starting point.

Now let q be the characteristic polynomial of whatever recurrence the entry asserts. The entry asserts it has order 75, so $\deg q = 75$, and it is monic. It holds from some index t on. From $\max(t, 7)$ on, the sequence satisfies both q and $q_{\min}$, and the minimal polynomial of that tail is $q_{\min}$ by the previous paragraph; therefore $q_{\min} \mid q$. Both are monic of degree 75, so $q = q_{\min}$. $\square$

5 Verification

Four checks, all in exact integer or rational arithmetic.

First, the walk model reproduces all 18 terms the entry publishes, exactly. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters the argument.

Second, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once over $\mathbb{Q}$ in exact rational arithmetic. Both return 75, and the exact run additionally confirms every coefficient is an integer — had any been a proper fraction the recurrence would not be the entry’s.

Third, the recovered recurrence was evaluated directly on the entry’s own published terms, with no matrices and no Berlekamp–Massey involved. It holds at each of the 0 indices where the entry publishes enough earlier terms to test it.

Fourth, the last coefficient was checked to be nonzero, which is what the identification above turns on; it is 1189085184.

Remark 4. The conjecture’s own statement was never read. The entry pins it down to the family of order-75 recurrences this sequence satisfies from some point on, and that family turns out to have exactly one member.

References

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